International Journal of Educational Research 119 (2023) 102183

Contents lists available at ScienceDirect

International Journal of Educational Research

journal homepage: www.elsevier.com/locate/ijedures

Quality and equity in the Peruvian education system: Do they progress similarly?

Rocío Serrano Rodríguez [a], Elisa Pérez Gracia [a,*], Lucy Puño-Quispe [b], Alejandra Hurtado-Mazeyra [b]

[a] Department of Education, Universidad de Córdoba, Córdoba, Spain
[b] Department of Education, Universidad Nacional de San Agustín de Arequipa, Arequipa, Perú

ARTICLE INFO

Keywords:
Socioeconomic status
Primary education
Student achievement
Equity
equal education
Educational quality

ABSTRACT

This study aims to understand the relationship between quality and equity in different regions of Peru. The academic results of students in Language and Mathematics in the second and fourth years of Primary Education from 2007 to 2018 were analysed. The results indicate improvements in the scores and a more homogeneous result distribution in the country, and show that area, poverty index, investment in education, permanence, and enrollment rates affect academic results and the coefficient of variation, and that the latter is also affected by subject and GDP. Lower scores are attributed to rural areas, with public centres presenting the greatest inequalities. Finally, slight increases in the socioeconomic index of families with fewer resources have an effect on score improvement.

1. Introduction

The main challenge facing educational policy is the maximization of educational excellence (quality) alongside the reduction of inequity (equity) (Pérez Dávila, 2018; Rodríguez, 2019; Sánchez & Manzanares, 2014; Villasmil et al., 2019). Education plays a key role not only in people's lives, but also in the economic development of the country to increase productivity and reduce inequity (Sulis et al., 2020).

Equity is understood as the constant search for equal educational opportunities, as a way to broaden the concept of equal treatment to include the academic potential and performance too. This concept is coherent with Rawls's second principle of justice, which establishes that there must be a fair equality of opportunities for social and economic progress, that is, it guarantees both the possibility of such change exists and that all people have a fair chance of achieving it (Fiske & Ladd, 2004; Guthrie & Wong, 2015). Therefore, equity does not only seek to eliminate inequality but also by the extent to which such gaps and elements of distribution of results are "fair" (Rolleston & Iyer, 2019). In this line, UNESCO (2018) recognizes that equity is the mean to ensure that distribution is fair and implies a normative judgment about distribution that indicates that perceptions of inequality vary depending on whether people consider educational circumstances to be 'unfair' or 'fair', and this must be carefully discussed to understand the given context. Discussing equity in education is challenging; due to systemic problems, inequality cannot be avoided. While equity has received considerable attention, its definition remains unclear, although there is a common understanding of the relationship between 'equity' and 'justice' (Gooden, 2015).

* Corresponding author.
E-mail address: elisa.perez@uco.es (E. Pérez Gracia).

https://doi.org/10.1016/j.ijer.2023.102183
Received 18 November 2022; Received in revised form 6 March 2023; Accepted 11 April 2023
Available online 14 April 2023
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On the other hand, educational quality must be understood as those directed processes, which provide an optimal scenario for the development of the teaching-learning process to achieve the purposes of education (Barba Miranda & Delgado Vadivieso, 2021) and as the sum of elements that make possible the achievement of the highest satisfaction and educational excellence (Huapaya Capcha, 2019). To do so, it is important to consider different aspects such as: development of science, cultural, political and economic aspects, and the proper functioning of the educational institution (Muñoz-Sánchez et al., 2022). In addition, educational quality should look upon the dynamic integration of some fundamental axes such as: institutional organization, human resources, school curriculum, educational didactics and academic results (Barba Miranda & Delgado Vadivieso, 2021). Furthermore, despite the fact that it is decisive to know how and how much students learn, there is another important key related to the way this learning is put into practice in different situations and contexts so as to analyse how it has an effect on the society to check whether it transcends the borders of the educational institution. Thus, it affects economic development, the productivity of people and the competitiveness of countries.

For all these reasons, it seems interesting to conceive the measurement of quality as proposed by Barba (2018) through a formula that includes a conception of the quality of education (QE) as follows: QE = ae, where "a" is the learning achieved and "e" stands for equity. Being the quality of education equal to learning, multiplied by the equity that exists in the educational institution and, through this, as a requirement, of equity in the social, economic, and political environment of education.

The need to provide a balanced response to these two demands (quality and equity) is materialized in constant debates across different educational systems and countries (Brito, 2019; Cardozo & Menese, 2019).

Currently there is a strong educational demand for educational quality manifested everywhere by an improvement in educational conditions (Guthrie & Wong, 2015; Ladd & Fiske, 2008). Expanding educational opportunities and improving quality are important facets of education. However, because economic inequality can reproduce itself in the form of educational inequality, attention also needs to be paid to the beneficiaries, to ensure that the Agenda for Sustainable Development 2030 is upheld, including its focus on "equity" Ogawa (2022).

Although the need for quality improvement should ideally be related to equality improvement, many debates on quality improvement focus on quality improvement for its own sake, to the exclusion of the perspective of how quality improvement can benefit all students. On the contrary, with the improvement of quality it can lead to the creation of new inequalities. To achieve a more equitable quality education for all, it is important not only to consider schools, but to focus on students as individuals (Ogawa, 2022).

However, few studies have addressed these two factors (Garira, 2020). The path to educational excellence must involve improving both students' academic results and equity (Kyriakides & Creemers, 2011; Kyriakides et al., 2019); as well as the promotion of meeting students' needs and individual growth (Huang & Chen, 2021). However, some authors have proposed that greater inequity may produce better academic results (Galaz & Arévalo, 2019; Parker et al., 2018). Kelly and Elliot (2018) and Rolleston and Iyer (2019) found that inequity may be associated with factors related to students' stability (socioeconomic status, gender, and ethnicity). Educational excellence has mostly been addressed in the literature by examining academic results as shown in standardized and international educational evaluations (Bastos, 2018; Brito, 2019; Zúñiga, 2018) and not by considering the equity levels of different geographical areas (Soo & Kyung, 2016; Charalambous et al., 2011; Gruijters & Berhrman, 2020).

Studies confirm that changes in school policy have an impact on student achievement, this study revealed the potential benefits of applying the Dynamic Approach to School Improvement (DASI) in promoting quality and equity over a longer period of time, therefore, longitudinal studies are needed to determine changes in the state of effectiveness of schools in terms of quality and equity (Kyriakides et al., 2019).

Nachbauer and Kyriakides (2019) conducted research on the indicators to measure equity and concluded that the lack of research on the subject may be due to the varied approaches and criteria used to address the issue. Therefore, there is a need to define the objectives of research on inequality from a methodological perspective. In contrast, Murillo and Martínez (2017) attribute this lack of research to the scarcity of prospective data and, therefore, recommend the development of longitudinal research that allows for long-term analyses of how these inequalities evolve. Troncoso (2019) adds that the phenomenon of academic performance and equity is highly complex and that the few existing longitudinal investigations on explanatory variables of equity come from developed economies, since the challenge becomes even more complex in developing countries (Garbanzo, 2014; Girotto, 2019; Ibáñez, 2017). Namukwaya and Kibirige (2019) also affirm that there is an absence of equity studies in developing countries, with studies in developed countries focusing on the improvement of academic results obtained in standardized tests and using their variance as a measure of equity (Checchi et al., 2014; Parker et al., 2018).

Hence, the importance of addressing the relationship of educational quality and equity (Huapaya Capcha, 2019); insofar as educational quality has the capacity to empower and provide tools to improve life and respond to the collective well-being of society (Barba Miranda & Delgado Vadivieso, 2021). Therefore, education is an indispensable part of the development equation, and has an intrinsic value that goes far beyond the economic dimension to provide people with the ability to decide their own destination. For this reason, the opportunity to receive education is key to human development (Barba Miranda & Delgado Vadivieso, 2021). Equity provides equal opportunities and the elimination of economic, social and political barriers that hinder the development of people. For this, equity and quality must be combined to be effective and extensive, it is necessary to provide a social value that favours the construction of successful educational practices for all (Cervantes et al., 2018).

2. Quality vs. equity in Latin America

There is a growing interest from the political scenario to raise the level of education quality (Jiménez & Jiménez, 2019). However, achieving this goal cannot come about without paying attention to education equity (Battistone et al., 2019), otherwise, excellent education would not be a reality. In this sense, the OECD (2018) points out that by supporting disadvantaged school contexts and

providing them with equal opportunities, education systems perform better and become stronger. Much research has shown that Latin America is marked by not only lower formative standards than other regions of the world but also a lack of equity in its academic results (Rivas & Sánchez, 2016; Trueba & Remuzgo, 2017; Waissbluth, 2019). Furthermore, studies such as those by Parker et al. (2018) and Galaz and Arévalo (2019) have shown that countries do not need inequity to achieve high performance. However, the results of Parker et al.'s study, involving a sample of twenty-seven countries, demonstrated that there is still a direct relationship between educational improvement and increasing inequality. This is consistent with the work of Friant and Sánchez (2018) and Battistone et al. (2019), who found that the trend toward quality is biased to the extent where it becomes inaccessible to the disadvantaged.

Many Latin American countries are marked by high socioeconomic inequality. Most low-performing students come from poor families, live in precarious conditions at home, are severely malnourished, and have little or no access to the health systems (Gruijters & Berhrman, 2020; Sánchez-Restrepo & Louçã, 2021). In addition, Rickles et al. (2019) study asserts that to contribute towards the improvement of equity and academic results, research must be aimed not only at visualizing what is happening in the different countries in Latin America and the Caribbean, but also within them.

It is generally assumed that the socioeconomic level of a student's family has a very early influence on the student's life, but this relationship is not homogeneous in all countries (Choi et al., 2018; Heckman, 2011). Along these lines, Helbling et al. (2019) point out that students' socioeconomic levels are key to the development of their academic performance and that differences in these levels have a greater impact on students' academic performance during the primary stages of education in developing countries (Raza et al., 2019; Sackett et al., 2009). Specifically, a study by Ortega et al. (2018) in Chile found that socioeconomic variables had a greater effect on students' performance in mathematics as their early educational stages are completed. The researchers affirm that these effects are greater in subjects such as mathematics that are generally taught in school and for which the contribution of the family and the environment is limited (Escobar et al., 2018; Zysset et al., 2018). Similar results were obtained by Cueto et al. (2016) in their study on the evolution of academic performance in Peru from 2000 to 2009; the smallest amount of academic progress was found in language subjects, in the early stages, and in students with a lower socioeconomic status. Despite this research background in Latin America, a recent study by Sulis et al. (2020) emphasizes the scarcity of studies in this area and the need to develop a more precise understanding of the association between families' socioeconomic levels and students' results in mathematics and language in order to contribute to the lowering of inequality and generate policies accordingly.

The selection of educational institutions by families also influences education excellence. Scientific evidence shows that there is a public-private school efficiency gap in Latinn America (Delprato & Antequera, 2022). In this sense, studies conducted by Gamboa and Waltenberg (2012) and by Ferrão et al. (2017) in the Latin American context incorporate the relationship between the type of school (public vs. private) students attend and the results students obtain, explaining that this variable is an important source of inequality given that the best results are found in private institutions. A study by Ganga Contreras et al. (2019) in different regions of Chile found that socially disadvantaged individuals tend to study in public schools, thus obtaining lower academic performance. However, a study by Castro et al. (2018) found that students from public centres with similar socioeconomic characteristics to the students from private centres tend to achieve similar academic results. It should be noted that in Latin America, the educational political trajectory is following a trend towards privatization compared to the rest of the world's educational systems (Díaz, 2019; Moschetti et al., 2019), despite the increase in inequality because of this privatization (Murillo & Martínez, 2017).

There is also scientific evidence for the differences in academic results between urban and rural areas. Recent studies by Cuenca and Urrutia (2019) and Johnes and Virmani (2020) identify a gap in the academic results observed in rural areas compared to those of urban areas. This was already evidenced by Guadalupe Mendizábal et al. (2017) through an examination of the results of Programme for International Student Assessment (PISA) tests (2012), through which they found a large gap between rural and urban schools in Peru, with this disparity being significant between regions and in favour of urban schools. As Bedoya and Cristancho (2018) indicate, in developing countries, rural schools receive the most economically disadvantaged students, while private urban schools receive students with the highest socioeconomic levels.

In addition to socioeconomic level, type of school, and area, other factors at the regional and/or national level may intervene in students' academic results. Investment in education, for example, has been addressed as a variable associated with the students' academic results in the early educational stages. In this sense, Cunningham et al. (2019) point out that investment in education is directly related to improvements in academic results; however, they emphasize the need to ensure that there is a better distribution of education investment among rural and public centres. In this regard, Fernández and Peréz (2016) state that investment in education in Latin America is higher than the average in OECD countries. In spite of this, Agasisti (2014) and Lorcu and Bolat (2015), show that these economic resources are not always in line with the real needs of the country's educational system, or the necessary control mechanisms that allow a real adjustment to these needs are not present. Hence, UNESCO (2017) states that the educational system of these countries is not becoming more equitable despite the economic efforts being made. Mazurek and Mielcová (2019) affirm that there is an association between Gross Domestic Product and investment in education, although this does not always imply an improvement in students' academic results (Gutiérrez et al., 2016).

Permanence rate and enrolment rate are the two other factors related to quality and equity that have been addressed in the literature. Regarding the former, Hall (2018) states that most Latin American countries present a low permanence rate in the early educational stages. Studies addressing enrolment rate and the improvement of academic results suggest that the increase in enrolment rates in the early school years is not directly related to equity (Valente, 2019).

3. The Peruvian context

Peru is one of the Latin American countries that has invested least in education in the last five years (Alldatanow, 2018), currently occupying position 33 out of 66 countries reported by UNESCO in its world report on education (UNESCO, 2019). Public investment in education in Peru is the responsibility of regional governments, which has generated a greater educational inequality between regions (Magallanes, 2016; Michón & Timaná, 2015) and manifested in a high variability of academic results within the country (MINEDU, 2018).

To address this inconsistency in academic results and improve the country's educational quality, Peru joined the international standardized evaluations designed by the Latin American Laboratory for Assessment of Educational Quality (LLECE, Spanish acronym). Consequently, a comparative study conducted in Latin America showed that Peru had a greater heterogeneity in results in the subjects of language and mathematics (TERCE, 2013). In parallel to Peru's participation in these international tests, it designed its own Census Evaluation of Students (ECE, Spanish acronym) which showed that the deficiencies in results extend to all institutions (urban vs. rural, public vs. private) in all courses and subjects evaluated (language and mathematics, mainly) (Cuenca & Urrutia, 2019). In other words, the inequities of the educational system are evident in the geographical distribution of students and according to the type of school management (public vs. private). While the former is relevant to the objective of fulfilling the educational needs of the population, the latter is influenced by students' families' ability to pay for their studies (Garret Vargas et al., 2017).

However, Peru has managed to achieve an enrolment rate of 80% - 85% in all regions, with the highest rates corresponding to the regions that implemented a more efficient method of public spending per student; this is in conjunction with lower dropout rates (Cruz-Melzi, 2017; Michón & Timaná, 2015; Paucar & Pretel, 2019). However, these attendance levels are still insufficient when compared to the educational systems of other developing countries (Reátegui et al., 2017).

The motivation for conducting the study has to do with the fact that Perú presents an interesting location for analysing educational progress, in terms of both quality (academic results in ECE tests) and the homogeneity of these results (equity) as well as identifying the economic and educational variables that explain such progress. We also analyze the effect of the socioeconomic index variable of each family on our two response variables (quality and equity). The analysis of the ways in which family socioeconomics relates to academic results and inequalities allows us to create a model that shows where the most significant improvements in performance and equity are obtained. This information is highly useful for the projection of educational policies.

In this way, the study aims to answer the following research questions:

Research Question 1: How have (1) the academic results in language and mathematics and (2) the equity in these two subjects evolved over the ten years of the study?

Research Question 2: Have there been any improvements in (1) the academic results in language and mathematics and (2) the equity of both subjects in a similar manner across the different regions of Peru during the ten years of the study?

Research Question 3: What variables (economic and educational) explain the variations (1) in the academic results and (2) in the equity in language and mathematics?

Being the goal of the present study is to understand the relationship between quality (measured through academic results in standardized regional tests) and equity (measured through the coefficient of variation) in different regions of Peru.

4. Methodology

4.1. Selected data and variables

The international databases resulting from the Census Evaluation of Students (ECE) of Primary Education of Peru corresponding to the period 2007-2018 were used in this study. ECE is a large-scale standardized test implemented by the Ministry of Education of Peru on a yearly basis through the Office of Measurement of Learning Quality (UMC, Spanish acronym).

The data used in the study were requested from the UMC of the Ministry of Education, indicating the purposes of their use and agreeing with the confidentiality protection statement.

The population is made up of students from both public and private schools in the region who are enrolled in the 2nd and 4th years of Primary Education. Only students who attend an educational institution with a ratio of less than five students were excluded. Therefore, it consists of non-probabilistic and convenience sampling technique. The data are grouped by region (n = 25; they correspond with all regions in the country) and academic year (second and fourth). A total of 26,471 educational centers participated (n. Private = 3,567 and n. Public = 22,904), with 9,985 located in urban areas and 16,486 in rural areas. The distribution of the sample by courses, periods and subjects is detailed in Table 1.

Table 1
Participating sample data.

Tests (ECE)	Period	n. Language	n. Mathematics
Second year	2007-2016 (10 years)	5051204	5056469
Fourth year	2016 and 2018 (2 years)	1016357	1015957

Source: UMC, 2019

Data from two dependent variables were obtained for use in this study: (1) academic results in Language and Mathematics (score) and (2) equity in the results of these two subjects (measured as the score's coefficient of variation). In other words, higher coefficient of variation (CV) values indicate greater inequality in the scores. For variable (1), the individual scores obtained in both the language and mathematics tests were considered, and for variable (2), the coefficient of variation in the individual scores obtained was calculated according to the region and academic year level. Both variables are organized by region (n= 25), academic year (second and fourth), and year (2007-2016 for the second year and 2016 and 2018 for the fourth year).

The following independent variables were used: (1) economic variables: gross domestic product (GDP), investment in education, socioeconomic index (SI) and poverty index;(2) educational variables: permanence rate, enrolment rate, type of management (public vs. private), subject (language vs. math), and area of the educational canter (urban vs. rural). These variables were obtained at the regional level and for each year of the studied period (2007-2018), except for the socioeconomic index, which was registered at the individual level, by type of management (public vs. private) and area (urban vs. public) in which the educational centre is located (Table 2).

4.2. Statistical analyses

Research Question 1: How have (1) the academic results in Language and Mathematics and (2) the equity in these two subjects evolved over the ten years of the study?

We performed a descriptive statistical analysis for both subjects (language and mathematics) at the student level (n = 6,067,561 for language and 6,072,426 for math) along the study period (2007-2018). We separately considered the second and fourth year. In these analyses, the mean and the coefficient of variation of the score attained for each subject (language of years 2 and 4 and mathematics of years 2 and 4) were calculated. In addition, two ANOVAs were performed to identify differences between years and subjects with respect to the score and the CV (only for the second course). In these ANOVAs, the year (10 levels), the subject (2 levels), and its interactions were included as factors in both models. A protocol for the exploration of data and residuals to assess normality and variance homogeneity was applied by following the recommendations of Zuur et al. (2010). The statistical significance of the p value was set at 0.05.

Research Question 2: Have there been any improvements in (1) the academic results in language and mathematics and (2) the equity of both subjects in a similar manner across the different regions of Peru during the ten years of the study?

A cluster analysis was performed to observe how the regions (n = 25) were grouped according to their mean scores (mean of both subjects in the second course in 2007 and 2016) and coefficient of variation (only for mathematics in the second course in 2007 and 2016). This technique allowed the different regions to be grouped based on the score and CV. First, the optimum number of natural groups of regions (K) was determined using a hierarchical clustering analysis; clusters were then generated with a "K-means" conglomerate analysis. Both methods were developed using SPSS version 20 for Windows (SPSSInc, Chicago IL). A hierarchical clustering analysis was performed using Ward's method, in which information is quantified as the sum of distances of each element regarding the centroid of the cluster to which it belongs. This was done by first calculating the mean vector for all the variables and the multivariate centroid for each cluster and then calculating the Euclidean distances between each element and the centroid (mean vector) of all the clusters. Finally, the distances for all the elements were combined. The "K-means" conglomerate method was used for cluster generation: "k" groups of regions were generated on the basis of their similarity to the score and CV.

Research Question 3: What variables (economic and educational) explain the variations (1) in the academic results and (2) in the equity in language and mathematics?

To determine which factors may influence the scores for the second course (Model 1) and the coefficient of variance for the second course (Model 2), two general linear mixed models (GLMM) were performed (only for second year, since only two years of data were available for the fourth course). In both models, the GDP by region, poverty index, investment in education, permanence, and enrolment rate were included as covariables, whereas subject (two levels), area (urban or rural), and management type (public or private) were added as fixed factors. In addition, the region was included as a random factor. A normal distribution and identity-link

Table 2
Description of the independent variables.

	Variables	Description	Type
Economic variables	Gross domestic product (GDP)	Wealth level of the region.	Ordinal
	Investment in education	Percentage of total government spending allocated to education.	Ordinal
	Poverty index	Statistical parameter defined by the UN to measure countries' standard of living compared to the human development index.	Ordinal
	Socioeconomic index (SI)	Index that characterizes the individual's family context according to a set of economic and social variables that combine information from the individual, the family, and the household.	Ordinal
Educational variables	Permanence rate	Degree of constancy, persistence, and dedication of the students. Continuity in the educational system.	Nominal
	Enrollment rate	Net rate of primary school enrollment of the population from 6 to 11 years old.	Ordinal
	Type of management	Public/private	Nominal
	Subject	Language and math	Nominal
	Area	Urban area (minimum of 100 grouped dwellings) and rural area (less than 100 grouped dwellings).	Nominal

Source: World Bank (World Bank Group), UNESCO's Institute of Statistics and INEI (National Institute of Statistics of Peru, Spanish acronym).

function were used for both models. For scores, the data is calculated at the individual level, while for CV, the data is calculated at the region level. In the case of the GLMM, each individual has certain characteristics (predictors), although the response variable is the same (CV) for all individuals in the same region. Therefore, the models are calculated at the individual level, assigning to all individuals in the same region the same CV value. The selection of the most plausible models was carried out by comparing Akaike's information criterion (AIC) in the models (Burnham & Anderson, 2002) following a backward procedure (Zuur et al., 2009). In particular, we compared the Akaike information criteria for small sample sizes (AICc value) in each candidate model and the best model (that with the lowest AICc). As a rule, a $\Delta i < 2$ suggests that the candidate model has a similar explanatory power to the a priori best model (Burnham & Anderson, 2002). We therefore selected all the models in which $\Delta i < 2$ with regard to the best model. Statistical analyses were performed by employing InfoStats software (Balzarini et al., 2002).

Socioeconomic index was not included in the model due to the lack of information for most of the years since it was only available for years 2015-18, but as Helbling et al. (2019) and Raza et al. (2019) point out, it is necessary to consider this variable when addressing variations in academic performance results. In the sense, to evaluate the effect of the socioeconomic index on the dependent variables (mathematics and language scores and coefficient of variation of mathematics and language), four non-linear regressions were performed. In these regressions, the socioeconomic index was the predictor variable. Sigmoidal equations were used for the scores, while exponential equations were used for the coefficients of variations.

Sigmoidal equations= pr1 + (pr2-pr1) / (1 + 10 ^ ((pr3-X1) * pr4))

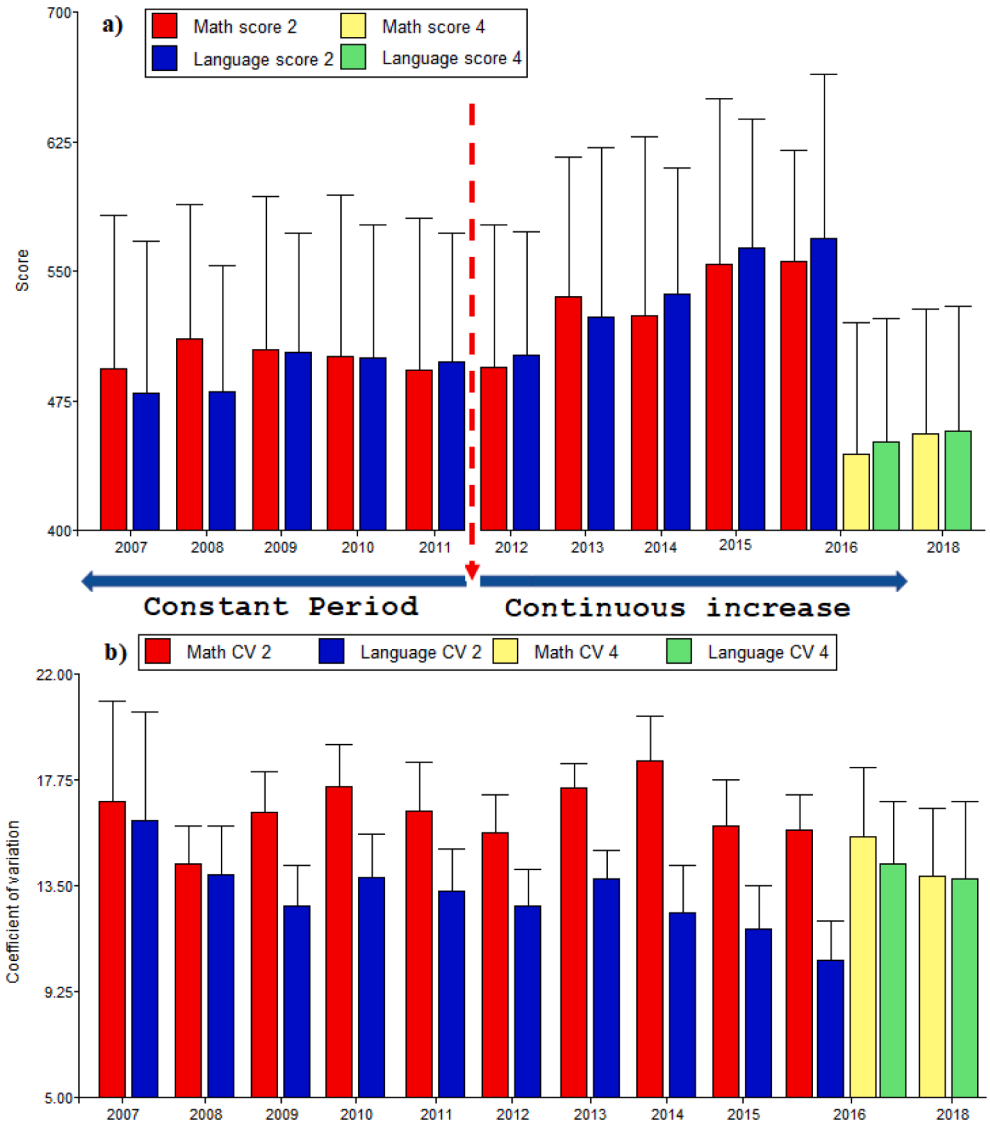

Fig. 1. The results of two subjects and the change in academic results throughout the period studied.

5. Results

The results obtained for each of the research questions are shown below.

Research Question 1: How have (1) the academic results in language and mathematics and (2) the equity in these two subjects evolved over the ten years of the study?

Regarding the descriptive results during the study period, the scores were (mean ± S.D. = 513.8 ± 91.8 and 519.8 ± 80.2 for mathematics and language in the second year and 443.9 ± 75.3 and 450.4 ± 71.3 for mathematics and language in the fourth year), while those for CV were (mean ± S.D. = 16.1 ± 2.7 and 13.1 ± 1.9 for mathematics and language in the second year and 15.5 ± 2.8 and 14.4 ± 2.5 for mathematics and language in the fourth year).

Fig. 1 shows the similarity between the two subjects and the change in academic results throughout the period studied (Two-way ANOVA, subject factor F = 0.09, df = 632718, p <0.7625, factor year F = 7217, df = 632718, p <0.0001). Similarly, the result of the interaction of the factors year and subject was also significant, meaning that the improvements in academic results are not parallel between the two subjects (two-way ANOVA, factor year * subject F = 545, df = 632718, p <0.0001). In this sense, two periods can be identified for the second year. The first of these is a period with stable academic results over five years and in both subjects (2007-2011), in which mathematics presents average values in academic performance superior to language. In the second period from 2012 to 2016, a progressive increase in the academic results of both subjects can be observed. However, in this second period, the average values in academic performance for both subjects are reversed with respect to the first period, as the language course has the highest values. For the fourth year, only the years 2016 and 2018 are available, an interval in which a slight increase in academic performance is observed in both subjects. On the other hand, in Fig. 1b, the CV values do not remain constant during the period studied, and the CV values for the two subjects are also different (two-way ANOVA, factor year F = 15248, df = 33632, p <0.0001, factor subject F = 145172, p <0.0001). Furthermore, the ANOVA results show a significant interaction between the two variables year and subject (two-way ANOVA, factor year * subject F = 30264, df = 33632, p <0.0001), which means that language and mathematics present an uneven evolution over time with respect to their CV (2007-2016). In mathematics, the values fluctuate throughout the study period without a clear trend. On the contrary, in language, there is a decrease in the values of the CV, that is, the results in the academic performance of this subject tend to be more homogeneous during the same period. In the fourth year, the values of the CV also decrease in both subjects between the two years for which we have data, which supposes a greater homogeneity in the academic results of this educational level.

Research Question 2: Have there been any improvements in (1) the academic results in language and mathematics and (2) the equity of both subjects in a similar manner across the different regions of Peru during the ten years of the study?

Due to the absence of significant differences in the academic performance for both subjects (see question 1), Fig. 2a shows the academic results of the joint average of language and mathematics in the different regions of Peru. However, for the CV with disparate behaviour between both subjects, the results are shown separately (Fig. 2b and c). Furthermore, in order to compare the results between regions at the beginning and end of the study period, the results are shown independently for each of these two years (2007 and 2016). Only the second-year data have been considered, since it is the only long time series available (10 years).

In Fig. 2a, we can see an increase in academic performance in 2016 compared to 2007 in all regions. Only four regions (Loreto, Ucayali, Huanuco, & San Martin) remained below the country average in both years. On the contrary, six regions remained above the country average in both 2007 and 2016 (Tacna, Moquegua, Callao, Arequipa, Junin, & Lima). Finally, the four regions that presented a greater increase in their yield in this interval of years were Ayacucho, Tacna, Huancavelica, and Cusco.

Further, the inequalities of the results corresponding to the language subject (2b) decreased from region to region, except in Cusco, San Martín, and Ancash, where they remained stable. It should be noted that in mathematics (2c), the differences between the regions decrease for the period studied.

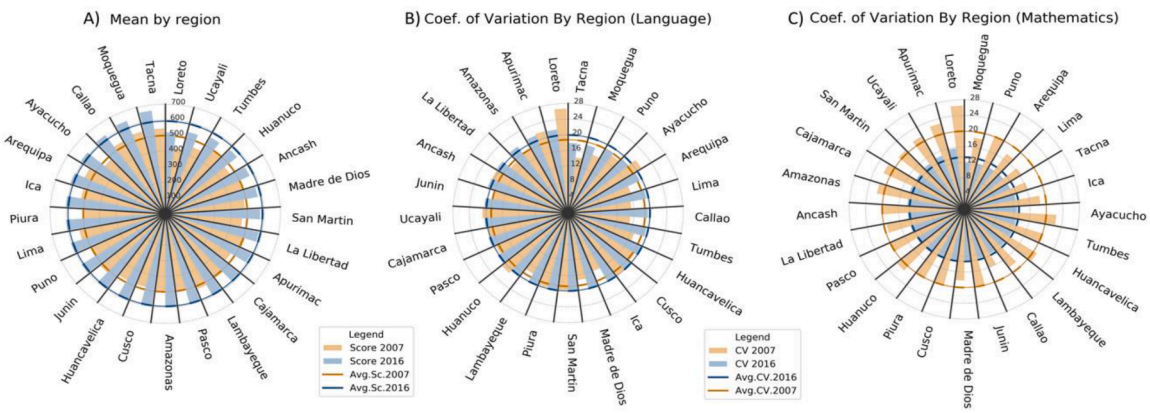

Fig. 2. Results of the joint average of language and mathematics in the different regions of Peru.

Four dendrograms were constructed to identify the similarity index existing between the 25 regions of Peru (Fig. 3). The first two dendrograms correspond to the joint average of both subjects in the academic results for the beginning (2007) and end (2016) of the study period. The other two dendrograms show only the results of the CV in mathematics, as it is the subject that presents the most variations (see question 1) for 2007 and 2016. The four dendrograms are for the second year.

In both cases (score and CV), dendrograms generate two separate groups of regions due to clearly different results between them. Regarding the score, both groups have a similar number of regions both in 2007 and 2016. However, the CV dendrograms show a different evolution for this variable, since the group with greater inequalities in the results (greater CV) included 16 regions in 2007, while this group only included 11 regions in 2016. The formation of these clusters largely reflects the proportion of rural/urban centers in each region, grouping regions with a higher percentage of rural centers in C1 (blue) and more urban centers in C2 (red). These results indicate that during the study period, there was a decrease in regional performance inequalities in mathematics.

Research Question 3: What variables (economic and educational) explain the variations (1) in the academic results and (2) in the equity in language and mathematics?

A GLMM was performed to determine which factors explain the variations in the score (Table 3: Model 1). The best model excluded two variables (type of management and GDP) where R2 was 0,22. The rest of the variables were retained in the best model and show a significant effect, except for the subject. The variable area (urban vs. rural) stands out, as students from urban areas have higher scores. Investment in education and permanence rate positively affected score, while the poverty index and enrollment rate negatively affected score (Table 3).

In the analysis of the variables that affect the results in the inequalities (CV: Model 2), the best model excluded two variables (investment in education and enrollment rate) where R2 was 0.39. The rest of the variables were retained in the model. These results show that subjects have a significant effect; mathematics presents more inequality than language. However, rural and public schools have higher levels of inequality in their academic results compared to urban and private schools. Finally, the poverty index and the permanence rate show a positive relationship with this inequality, while the GDP shows a negative effect, this is less inequality of academic results to higher GDP (Table 3).

Finally, the effect of the socioeconomic index was not included in the model because we only have data for the years 2015-2018, but we did consider it important to address this effect independently. In this sense, the socioeconomic index (SI) of the families showed a marked effect on the two response variables, both in the academic result (score) and in the inequalities in the results of language and mathematics (CV).

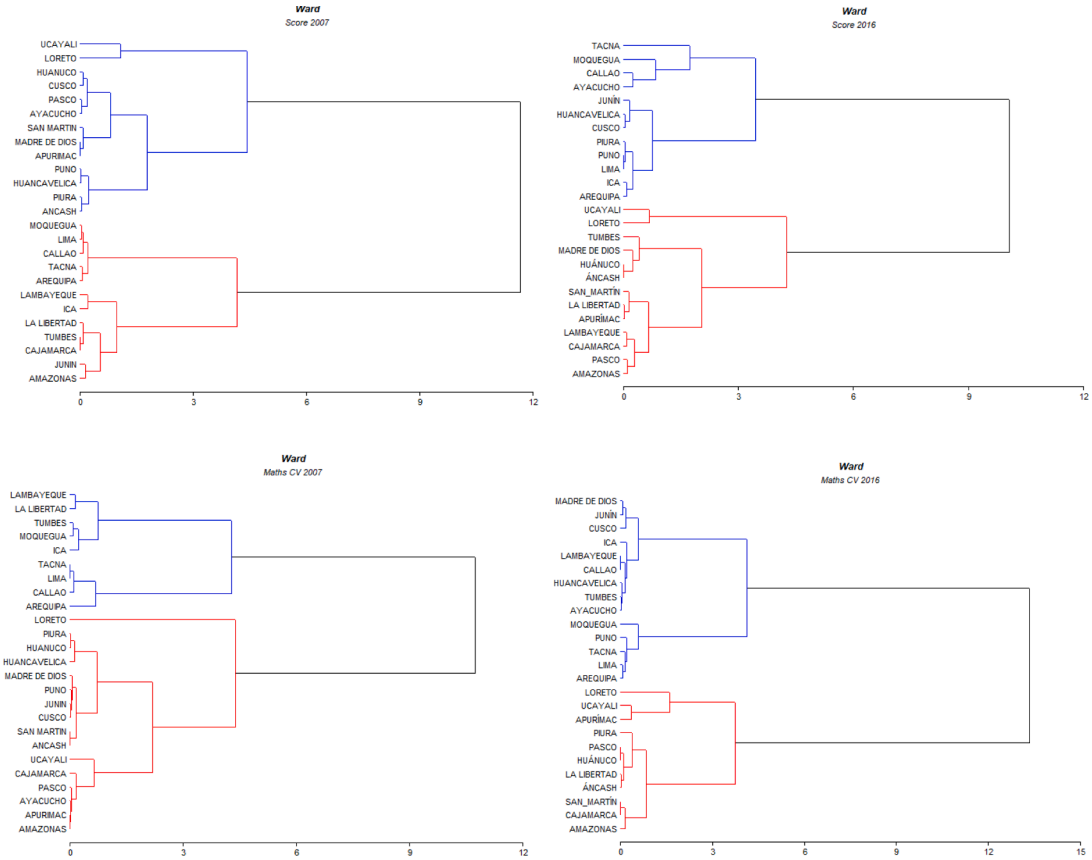

Fig. 3. Four dendrograms to identify the similarity index existing between the 25 regions of Peru.

Table 3
F-value, p-value and coefficients of the variables retained in the best models to explain score (model 1), and CV (model 2).

Variable	*df*	F-value	*p*-value	Coefficients ± S.E.
Model 1: Score				
Intercept	464022	4174.8	<0.0001	589 ± 9.3
Area		6017.1	<0.0001	Urban = 21.2 ± 0.27
Poverty index		9.4	0.002	-0.16 ± 0.05
Investment in education		1953.5	<0.0001	0.02 ± 0.003
Permanence rate		376.9	<0.0001	2.62 ± 0.13
Enrollment rate		524.5	<0.0001	-3.87 ± 0.17
Subject		0.7	0.4	Math = 0.2 ± 0.2
Model 2: CV				
(Intercept)	464146	3779.6	<0.0001	6.9 ± 0.13
Subject		237600	<0.0001	Math = 2.9 ± 0.01
Type of management		1913.8	<0.0001	Private = -0.39 ± 0.01
Area		7070.9	<0.0001	Urban = -0.59 ± 0.01
Poverty index		109045	<0.0001	0.08 ± 0.004
GDP		112.4	<0.0001	-0.002 ± 0.0001
Permanence rate		1095.9	<0.0001	0.05 ± 0.001

Coefficients for the level of fixed factors were calculated using reference values of 'public' in the variable 'type of management', 'rural' in the variable 'area' and 'language' in the variable 'subject'.

The relationship between the SI and the academic results adjusts to a sigmoid model in which we can distinguish three sections and is common to the two subjects studied (Fig. 4). In the first section, the academic results grow exponentially as the SI of the families increases until they reach the maximum slope when the SI value is -1.24 (for both subjects), which corresponds approximately to the third lower than SI values. From then on, the yield curve can stabilize progressively as the SI increases, which corresponds to its intermediate values. In the last section, corresponding to the families with the highest SI, the performance values stabilize around an asymptotic value of 524 and 502 for language and mathematics, respectively.

At the same time, the values of the inequalities in the academic results follow a negative exponential model when they are related to the SI. For the lower third of the SI values, these inequalities decrease with a maximum slope, which means that in this section of the SI, a slight increase in SI corresponds to an exponential decrease in the inequalities of the results. The slope of the CV curve gradually decreases as the SI increases, although the mathematics curve maintains a greater slope than the language curve, which means that the differences between results are minimized more quickly in mathematics than in language by increasing families' SI.

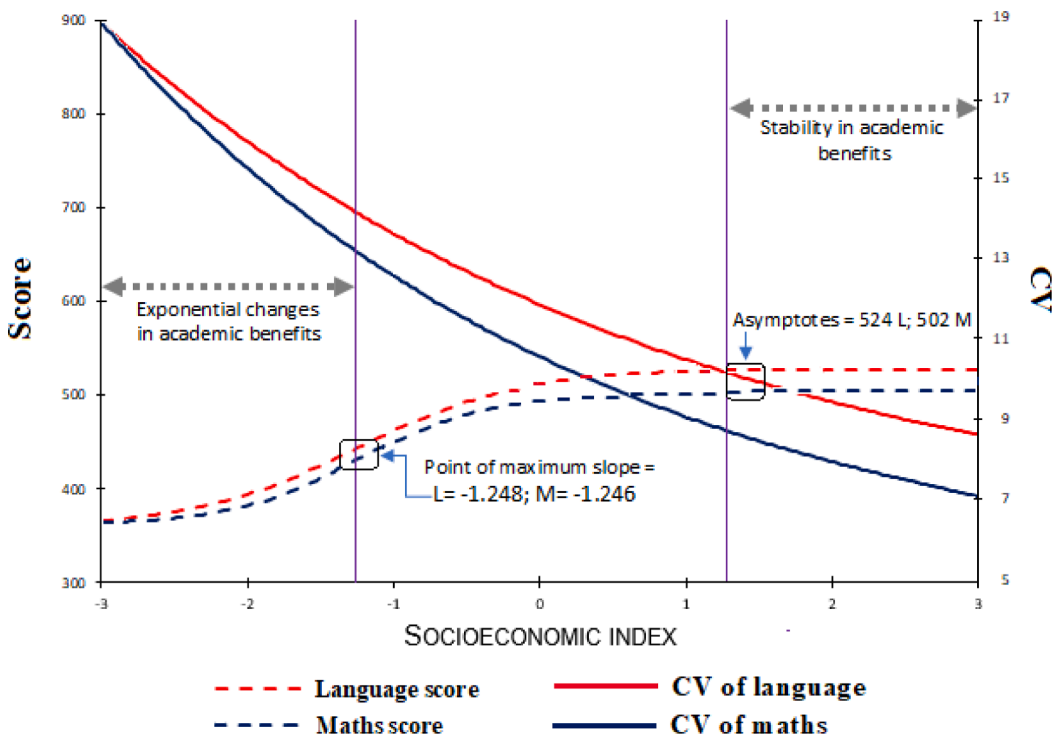

Fig. 4. The relationship between the SI and the academic results.

6. Discussion

Educational systems have mainly been analysed by examining the results of standardized exams (Batos, 2018; Brito, 2019; Zúñiga, 2018). Most educational authorities around the world are focused not only on increasing students' scores in different subjects but also on reducing inequality (Rodríguez, 2019; Rolleston & Iyer, 2019; Villasmil et al., 2019). However, the heterogeneity of students' scores has received little attention in the literature, as many researchers solely focus on academic results (Garira, 2020; Sulis et al., 2020).

Interestingly, our results show that both score (quality) and CV (equity) show complementary information to analyse the evolution of educational performance in different regions of Peru, so it contributes to the attention made by Gruijters and Berhrman (2020) of considering equity among excellence education, and not only quality. Moreover, it also agrees with the perspective of Huapaya Capcha (2019) who directly connect and evince the need to approach both quality and equity together.

Therefore, while the topic is little explored in the literature (Nachbauer & Kyriakides, 2019; Namukwaya & Kibirige, 2019), we have found that academic results and inequalities among these results do not evolve in parallel. We have demonstrated that in Peru, the language subject is different from the mathematics subject in terms of CV. However, in terms of score, both subjects evolved similarly, with an increase in academic performance from 2012. As Garbanzo (2014), Girotto (2019) and Troncoso (2019), Ibáñez (2017) argue, the evolution of these two elements in Latin America (academic results and equity) is highly complex, even though greater inequity may be related to improvements in academic results (Parker et al., 2018). In light of our results, we will discuss the different questions proposed in this paper.

Research Question 1: How have (1) the academic results in language and mathematics and (2) the equity in these two subjects evolved over the ten years of the study?

Fig. 1a and b show that on the one hand, academic results have tended to improve over the past five years, while the results have become more homogeneous in the country, at least for the second-year language subject. Regarding the fourth year, although there were only two years of data available, there is a similar trend with a slight improvement in performance as well as a decrease in the heterogeneity of academic results. In this sense and as anticipated by Pérez Dávila (2018) and Villasmil et al. (2019), Sánchez and Manzanares (2014), the educational policy in Peru seems to be meeting the challenge of maximizing educational excellence and reducing inequity, although the scale of the effect is not yet at the desirable level. In this sense, it could be appropriate to review how educational policies evolve in this context paying specific attention to education finance (Guthrie & Wong, 2015).

Overall, a period of stagnation in the primary education system can be identified between 2007-2011 with a clear improvement in the most recent five years of the study. However, as indicated by authors such as Parker et al. (2018) and Galaz and Arévalo (2019), there have been improvements in the academic results of the two subjects studied, but not in the homogeneity of the results obtained in mathematics when compared with the results obtained in the language CV. The average academic improvements in mathematics present dispersion values fifty-percent greater than those of language. Therefore, the difficulty of balancing quality and equity that Cardoso and Menese (2019) and Brito (2019) manifest in their studies on educational inequalities in developing countries becomes evident.

In this sense, the improvements in the average results are not a generalized effect but are rather caused by a certain fraction of students that improved their performance while the rest stagnated. This is the reason for the high CV values in mathematics that did not decrease throughout the entire studied period. As Friant and Sánchez (2018) and Battistone et al. (2019) affirm in their studies, quality is biased in one sense, making it inaccessible to the most disadvantaged (Garret et al., 2017). Accordingly, Escobar et al. (2018) and Zysset et al. (2018) confirm that the effects are greater in subjects such as mathematics which, due to their nature, require the resources and context of the school, with a more limited contribution of the family and the immediate environment.

Research Question 2: Have there been any improvements in (1) the academic results in language and mathematics and (2) the equity of both subjects in a similar manner across the different regions of Peru during the ten years of the study?

According to Soo and Kyung (2016), Charalambus et al. (2019), and Rickles et al. (2019), when we conduct an in-depth analysis of a country, we can see notable differences within it. In this sense, the dendrograms show that there are two clear regional groups, both in the score and in the CV, indicating that regions tend to group in accordance with the proportion of rural vs. urban centres. However, although these two groups are maintained throughout the studied period, improvements in the results have been observed in both, which means that the educational gap does not widen, unlike the findings of Bedoya and Cristancho (2018), Cuenca and Urritia (2019) and Johnes and Virmani (2020). Further, the information provided by the CV group indicates that the group of regions with the highest proportion of rural centres has considerably reduced its inequality, from 16 to 11 regions.

However, when we analyse the five regions that became part of the group with the least CV in 2016, a decrease in the number of public educational centres and an increase in the number of private centres can be observed in all of them. As anticipated by Díaz (2019) and Moschetti et al. (2019), in Latin America, educational centres are increasingly being privatized; this variable is an important source of inequality (Ferrão et al., 2017; Gamboa & Waltenberg, 2012; Murillo & Martínez, 2017).

Research Question 3: What variables (economic and educational) explain the variations (1) in the academic results and (2) in the equity in language and mathematics?

The explanatory variables model of the score results shows that large variations in the results can be attributed first to the rural/urban nature of the centres (educational variable), and second to the investment in education (economic variable). These factors are highly interrelated because the best educational infrastructure is usually concentrated in urban areas and private centres (Cunningham et al., 2019), although it should be noted that Peru is one of the Latin American countries that invests least in education (Alldatanow, 2018; UNESCO, 2019). In line with the findings of Guadalupe Mendizábal et al. (2017) and Cuenca and Urritia (2019), this study shows that rural areas present the lowest academic results in both subjects compared to urban areas, generating a great variability in academic results within the country (MINEDU, 2018). Thus, these results also coincide with the study carried out by Delprato

and Antequera (2021) who clearly evinced the positive efficiency gap for private schools.

As previously indicated, dropout rates are high in the Latin American educational setting; in this case, permanence rates have increased, which has been accompanied by an improvement in academic results (Paucar & Pretel, 2019). On the other hand, in the explanatory variables model of the CV, the greatest inequalities in the results occur in the subject of mathematics, in public centres, and in rural areas (Rivas & Sánchez, 2016; Trueba & Remuzgo, 2017; Waissbluth, 2019). As Ortega et al. (2018) indicate, certain subjects, such as mathematics, depend more on the educational context than on the family context, with the best results corresponding to students in private education settings in the Latin American context (Ferrão et al., 2017; Gamboa & Waltenberg, 2012; Ganga Contreras et al., 2019). Our results are consistent with those cited above as well as with the findings of authors such as Heckman (2011) and Choi et al. (2018). However, other developing countries with a high percentage of rural population but with heavy investment in education do not present such educational imbalances (Raza et al., 2019; Sackett et al., 2009). Another explanatory educational variable of the CV was the permanence rate. In this regard, Hall (2018) explained that in Latin America, there is a low permanence rate in the early educational stages; however, Peru has managed to improve its enrolment rate, increasing it to 85%, although this is not directly related to the improvement in inequality (Valente, 2019) and it is not considered to be sufficient when compared to other Latin American systems in which this rate is 99% (Reátegui et al., 2017).

Regarding the economic variable, GDP was an explanatory factor for the CV in Peru. Our results are in accordance with the UNESCO report (2017) and the research of Agasisti (2014) and Lorcu and Bolat (2015), which state that an educational system does not become more equitable simply through economic investment if such investment does not address the country's real educational needs. Countries with a higher poverty index will see their CV increase. It also agrees with the statement made by the OECD (2018) to make countries aware of the great importance of promoting equity to strengthen education systems. In addition, it aligns with the Agenda for Sustainable Development 2030 and the appeal made by Ogawa (2022) to consider equity as the focus of educational goals.

Helbling et al. (2019) longitudinal work with primary education students examined the effects of families' socioeconomic levels for the same two subjects examined in our study, with the conclusion that the SI is a key factor in the development of students' academic results, which is particularly true in the early stages of education. The relationship between SI and score identified in the present study shows the exponential effect of the SI on the score for families with a lower SI. However, our results show that once families reach a given SI threshold, the effect of SI on the score dramatically decreases. Interestingly, our results also show that the dissimilarity of the academic results (CV) decreases at their maximum rate as the SI increases when the families are within the lower SI rank. On the other end of the SI, the scores of students from families with higher economical levels do not increase, suggesting that there is a threshold in the effect of the SI on students' scores. However, the dissimilarities among results (CV) still continue to decrease along the SI rank, although with a lower slope than that at lower SI.

As previously mentioned, the differences between the CV in math and language increase along the SI rank, which may be due to differences in the learning requirements of both subjects. As explained by Malmberg and Sammons (2018), in mathematics, the differences in the results are more persistent in the period studied when compared to those for language, which is more sensitive to students' family situation and environment. In this sense, it is necessary to further investigate the educational context to understand this variation in academic results; or as Kelly and Elliot (2018) indicate, explore other stability factors for students (gender and ethnicity) to determine their effect on these variations.

7. Conclusion

The variation in academic results in Peru makes the country a particularly interesting context for this study, which addresses three of the issues that have been identified as research gaps in previous studies: i) analysing the evolution of academic results and equity by examining prospective and longitudinal academic results (Murillo & Martínez, 2017); ii) addressing the two factors of quality and equity in a Latin American context (Garbanzo, 2014; Girotto, 2019; Ibáñez, 2017; Troncoso, 2019); and iii) exploring the effects of educational and economic variables on academic results as well as inequalities in these results (Garira, 2020; Sulis et al., 2020).

Our results show that educational systems should not be analysed solely by examining academic results; it is also necessary to examine their homogeneity (Kyriakides et al., 2020). Policies must be aimed at developing educational systems that advocate educational excellence while reducing inequity. In addition, focusing on low-performing actors may be critical to the success of educational policy interventions in the Latin American context.

In this regard, improving the conditions of the lowest socioeconomic strata will have the greatest impact on academic performance and inequality. There is an urgent need to overcome this inequity recently highlighted by Sánchez-Restrepo and Louça (2021), there should be an alternative to promote low performing among students who belong to low socio-economic class. In practical terms, the SI results suggest that there are well-identified minimum thresholds below which an increase in economic efforts is exponentially rewarded in terms of academic improvements (score) and decrease in the inequalities of these results (CV). The differences between mathematics and language in the CV as the SI increases suggest that in the case of mathematics, the results tend to be more homogeneous (less CV) than in the case of language. Moreover, in this context, the debate between private and public education as well as the neoliberal educational policies must be address in order to enhance equity (Delprato & Antequera, 2021).

As Troncoso (2019) and Murillo and Martínez (2017) point out, the phenomenon of academic results is highly complex, and it is important to develop longitudinal studies that cover the primary and secondary stages of education, which would allow an analysis of all the relevant variables. Most studies in this field have focused on a specific year and attributed academic results to a specific moment in time. As has been highlighted in literature reviews in Latin America, more longitudinal evidence is needed to understand the effect of schools on the academic results of students over time.

It is obvious that to achieve equity and provide quality primary education for all, greater attention must be paid to the inclusion of

students from lower socioeconomic backgrounds. Differences in family income in the primary stage are a much more significant predictors of academic achievement than differences in income during adolescence. Our research shows that lower socioeconomic classes are more sensitive to increased performance when their economic situation is slightly improved. Thus, there is a point at which students' academic results remain steady despite their families' economic improvement.

It would also be interesting to determine whether, in addition to the economic and educational variables that have been introduced in this article, other factors may be interfering; as suggested by Kelly and Elliot (2018), students' stability factors (gender and ethnicity) should be examined.

As can be concluded from our results, equity has yet to be achieved among the different regions of Latin America, and current policies must be reconsidered to close the gaps between different socioeconomic contexts.

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